

AUTOMATED TOPIC MODELLING OF ACADEMIC PAPERS USING BERTOPIC

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Statement of Originality

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Abstract

This study investigates the use of automated topic modelling to identify and analyse research themes within the field of Creativity Support Tools (CST) Evaluation. BERTopic is applied to the full text of 169 academic papers to identify groups of semantically similar documents and explore the associated topics and keywords. The resulting topics are manually interpreted and validated, and the findings are compared with the manual thematic analysis of the same papers conducted by Cox et al. (2025).

The comparison examines the extent to which automated topic modelling can identify themes similar to those found through manual analysis. The findings show that BERTopic can identify several coherent and recognisable groups within the literature, but that these clusters do not necessarily correspond to the analytical themes identified by Cox et al. This is because BERTopic primarily groups papers according to similarities in their textual and semantic representations, which may reflect their subject, terminology, or application domain rather than the specific research focus.

Overall, BERTopic is shown to be useful as an exploratory and screening tool that can help detect patterns, organise and reduce a large literature corpus, but human interpretation remains necessary to determine the relevance and meaning of the identified themes.

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1. Introduction

1.1 Background

The growing number of published papers makes it increasingly difficult for researchers to review, organise, and identify key themes across large collections of literature. Traditional thematic analysis typically involves manually reading and coding papers, identifying recurring patterns, and grouping these into broader themes, often with multiple researchers independently reviewing the data before consolidating a final set of themes (Clarke and Braun., 2017). While effective, this process is time consuming and difficult to scale as the volume of literature grows.

Topic modelling offers a potential solution to this challenge. It is an unsupervised machine learning approach designed to systematically analyse corpora and identify latent themes within the data.

Historically, topic modelling has predominantly used Latent Dirichlet Allocation (LDA), first introduced by Blei et al. (2003). This method is effective at finding latent thematic structures but struggles with the complex meanings and context found in highly technical and academic writing. This is largely because LDA represents text as a 'bag-of-words', treating words as independent of their context. Due to this, the model can fail to distinguish between different meanings of the same word, which limits its ability to capture the semantic nuances of the text.

More recent approaches use neural and transformer based language models to capture richer contextual representations of text. BERT (Devlin et al., 2019), for example, is a transformer based language model that generates contextual representations of words and text based on their surrounding linguistic context. Unlike traditional 'bag-of-words' representations, these embeddings encode semantic relationships between words and documents within a high dimensional vector space.

BERTopic (Grootendorst, 2022) builds on this approach by using a transformer based document embeddings as well as dimensionality reduction and clustering to identify groups of semantically similar documents/papers. The generated clusters are then represented using class based TF-IDF method to identify words that are distinctive from each topic. This allows BERTopic to take a more semantic approach, rather than relying on the word co-occurrence.

However, the use of contextual embeddings also raises some potential limitations. Because transformer models are trained on large real world corpora, they can encode existing societal and linguistic biases, which may influence the relationships between concepts represented in the vector space. A separate limitation is that general semantic representations may not fully capture domain specific meanings. For example, within the CST literature, the term 'evaluation' may refer to particular methodological approaches, evaluation criteria, user studies, creativity measures, or other context specific concepts. Although the model may recognise the general semantic meaning of 'evaluation', it may not necessarily distinguish these domain specific uses in the way a researcher familiar with the literature would. This represents a limitation when using general purpose language models to analyse specialised academic corpora as they may not capture those specific meanings.

Moreover, unlike LDA, BERTopic does not require the number of topics to be specified in advance. Instead, the number of topics emerges from the clustering process and is influenced by parameters such as the minimum cluster size and the characteristics of the dataset.

Therefore, this project focuses primarily on BERTopic because of its ability to incorporate contextual embeddings and identify clusters of semantically similar documents. Although BERTopic has been widely applied to texts such as abstracts, reviews, news articles, and online content, its application to full text academic papers remains comparatively limited.

1.2 Objectives and Research question

This project will apply BERTopic to a corpus of papers drawn from Cox et al.'s (2025) systematic review of Creative Support Tools (CSTs) and will evaluate the extent to which the resulting topics correspond to the themes identified manually in that review.

- To apply BERTopic to the corpus and identify its latent topic structure.
- To evaluate the extent to which the topics identified by BERTopic align with the themes reported by Cox et al. (2025) (User experience with CST, Creative Artefact quality, and User Centric Benefits).
- To critically evaluate the limitations of using an automated, domain agnostic clustering method to approximate a manually coded, outcome measure based classification.

Research Questions:

1. How does the thematic structure produced by BERTopic compare with the manual thematic analysis presented in Cox et al.(2025)?
2. To what extent can BERTopic identify patterns consistent with the findings of Cox et Al. (2025), given that the two approaches organise papers differently (application domain vs. outcome measure)?

1.3 Cox et al. (2025) as Benchmark

Cox et al. (2025), *'Beyond Productivity: Rethinking the Impact of Creativity Support Tools,'* conducted a ten year review of 172 CST evaluation papers, manually coding each paper's reported outcome measures into three overarching themes:

- User Experience with CST: measures of the user's experience during the ideation process while interacting with the CST.
- Creative Artefact Quality: measures of the quality of the creative output produced (e.g. writing, music, design).
- User Centric Benefits: measures of how the CST benefited the user beyond the interaction itself, such as effects on wellbeing or confidence.

However, these themes are not mutually exclusive, meaning that a single paper could report measures from more than one theme. So, the outcome percentages reflect the proportion of papers using each type of measure. With 90% for user experience, 54% for Creative Artifact Quality and 15% for User Centric Benefits.

From the three themes, User Centric Benefits had the lowest score, the authors argue that this is due to not having enough research on this theme as past researches have neglected the benefits CST might have in the user's life outside of using the tool.

2. Literature Review

2.1 AI-assisted Thematic Analysis:

In thematic analysis it is essential to comprehend that codes and themes are products of the researcher's judgement instead of a statistical result. Coding involves capturing the reason a segment of the data is relevant to the analysis. Saunders et al. (2023) argue that a well formed code should represent a 'complete thought' rather than a simple categorical label, because a label itself just groups the data according to the surface similarities of the words without actually capturing the researcher's interpretation of the underlying meaning.

Developing themes takes this process a step further. Braun and Clarke (2006) emphasise that the importance of a theme should not be determined only by the frequency of the words throughout the text, but also by its 'keyness', the extent to which it is relevant to the research question.

This distinction is key to understanding the relationship between thematic analysis and automated computational approaches. This is because computationally identified topics might group together documents according to textual/semantic similarities, whereas a researcher identified theme represents an interpretation of the data in relation to a particular analytical purpose.

2.2 AI and LLMs as assistance:

Recent methodological research has started to look at how large language models (LLMs) can be used to support researchers during thematic analysis rather than replacing them. Goyanes et al. (2025), for example, propose a six phase ChatGPT assisted protocol covering different stages of the analysis, where it is used to help identify patterns, extract convergences and divergences, and find supporting quotations, while the researcher is still responsible for checking the results and producing the final synthesis. This shows that LLMs can be useful for reducing some of the manual work involved when working with large amounts of qualitative data.

Naeem et al. (2025) take a similar approach but place more emphasis on how the researcher directs the analysis. Their method uses structured prompts to guide ChatGPT through keyword selection, coding, theme development, and conceptualisation. The model is also given the research questions and theoretical framework before the analysis takes place. This approach is significant because it means that the AI is not simply asked to find whatever patterns are most obvious in the data. Instead, the analysis is guided by what the researcher is actually trying to investigate. The researcher therefore still has an important role in deciding what is relevant, while the LLM can help with processing and organising the data.

There are also indications that LLMs may not be equally effective at all stages of thematic analysis. Morgan (2023) compared ChatGPT's output with his own previous manual analyses of two datasets. He found that the LLM was better at identifying more clear and descriptive content, such as surface

level topics and the main elements, but was less successful at identifying the more subtle interpretive themes produced through manual reflexive analysis. In other words, the model could identify things that were clearly present in the data, but had more difficulty identifying the broader idea or meaning connecting them.

Wiebe et al. (2025) provide another example of this distinction through their Hybrid 'Human-in-the-Loop' Thematic Analysis (HHitL-TA) framework. They divide the different stages of thematic analysis between the researcher and the LLM. Researchers establish theory driven constructs and the final codebook, while the LLM proposes codes and themes across the dataset. The outputs are then reviewed by the researcher before they are accepted, merged, or discarded. Their case study also found differences between the patterns identified through different analytical approaches, suggesting that automated analysis does not necessarily produce the same results as more fine grained human reviewed analysis.

Overall, these studies suggest that LLMs can be useful for reducing the amount of manual work involved in thematic analysis, but they do not remove the need for researcher interpretation. They can help identify potential patterns, organise information, and extract relevant evidence, while the researcher remains responsible for deciding why a pattern matters and whether it is relevant to the research question. This supports the idea of using AI as an assistance tool within thematic analysis instead of completely erasing the manual analysis by the researcher.

Another consideration is the propensity of LLMs to hallucinate. Researchers should therefore consistently request supporting evidence from the source material when interacting with an LLM, in order to verify and corroborate any claims it generates.

2.3 Topic Modelling and Thematic Analysis:

The distinction between automated pattern detection and researcher interpretation also applies to topic modelling. Although topic modelling and thematic analysis can both be used to identify patterns within a collection of text, they do not necessarily identify these patterns in the same way or for the same purpose. Gillies et al. (2022), for example, discuss how topic models can identify patterns within a corpus based on relationships between words, but these patterns do not necessarily correspond to the themes that a researcher would identify through qualitative interpretation. Rodriguez and Storer (2020) describe this difference in terms of lexical and compositional semantics. They argue that topic models focus on relationships between words and their occurrence in the data, whereas qualitative analysis considers how words and ideas come together to form a broader interpretation of a phenomenon. This means that a topic model may group documents that discuss similar subjects without necessarily capturing the underlying analytical meaning that a researcher is interested in.

Despite these differences, topic modelling and qualitative thematic analysis are not completely separate approaches. Rodriguez and Storer (2020) found that several topics identified by their model corresponded with major themes identified through qualitative coding of the same dataset. Similarly, Miner et al. (2023) compared LDA topics with codes from a previous qualitative study and found that various topics were strongly associated with specific human generated codes. The strongest correspondence was found for more concrete codes, such as family leave, medical training, and breastfeeding/pumping. These findings suggest that, in some cases, an unsupervised topic model can identify categories that are recognisably similar to those produced through human coding.

However, this correspondence becomes weaker when the qualitative analysis involves more abstract or interpretive concepts. Miner et al. (2023) found that topics associated with concepts such as ‘Power’, ‘Agency’, and ‘Respect’ showed weaker relationships with the corresponding qualitative codes, while some human generated codes were not clearly represented by any topic. This reveals that topic modelling may be better at identifying concrete or explicitly discussed subjects throughout the text than broader analytical concepts that are important but rarely observed in the text. Gillies et al. (2022) similarly argue that topic models can help identify what a collection of documents tends to discuss, but this does not necessarily mean that they identify ‘how’ or ‘why’ the documents discuss those subjects in relation to a particular research question.

This distinction is significant for the present study because the themes identified by Cox et al. (2025) are on a specific analytical focus: ‘the outcome measures used to evaluate Creative Support Tools’. A paper may therefore be grouped by BERTopic because it discusses a particular technology, user group, or application area, while Cox et al. may classify the same paper according to the type of outcome measure it reports. The two approaches can therefore produce different representations of the same literature without either necessarily being incorrect.

This raises an important question for the use of BERTopic in literature thematic analysis: whether an embedding based topic model can reproduce themes constructed according to a specific analytical focus, or whether it mainly recovers the general topical structure of the literature. This question provides the basis for examining the extent to which the BERTopic clusters in this study correspond to the manually constructed themes reported by Cox et al. (2025).

2.4 BERTopic and existing applications:

Studies in applying BERTopic to full text papers are limited, the method is most commonly used on abstracts, online reviews, social media posts, and news articles.

An example of this is the use of BERTopic in Lezhnina’s (2023) study , which examined mental health related literature in academia to extract insights from the field. In contrast to my study, she worked with 2,846 abstracts rather than full text. This produced a 27 topic model with 21% of documents classified as outliers, which she treated as a successful and interpretable result. The topics 'internet addiction' and 'psychometrics of depression' had the highest classification confidence scores (0.998 and 0.995, respectively), indicating that her custom labels closely matched the underlying topic content.

However, she did not apply any text preprocessing to her abstracts, stating that ‘BERTopic does not require data preprocessing.’ Her reasoning was that domain specific terms were already interpretable to specialists, and that BERTopic, unlike LDA, does not rely on a ‘bag-of-words’ approach, meaning such cleaning is not necessary to produce coherent topics.

While this may be defensible for studies working only with abstracts, my study applies topic modeling to 169 full text papers, which contains a lot of unnecessary content that abstracts do not (e.g., references, headers, funding statements, figure captions).Without preprocessing, this content would introduce considerable noise, degrading the quality of the resulting HDBSCAN clusters. In practice,

the embedding model and BERTopic pipeline could not be expected to perform reliably without this cleaning step. Therefore, my study incorporates both preprocessing and hyperparameter tuning to address these challenges.

2.5 BERTopic + LLM

An example of a study that integrates an LLM into BERTopic is Ma et al. (2025). The authors compared LDA and BERTopic using 1,837 abstracts about opioid related cardiovascular risks in women. ChatGPT-4-Turbo was used to interpret the results of both models. In BERTopic, the LLM was directly integrated into the pipeline to create one sentence labels for each topic. For LDA, the researchers manually provided the LLM with the 100 most important words for each topic, their probabilities, and the grouped abstracts, allowing the model to generate similar topic descriptions.

The topic descriptions produced by the LLM for LDA and BERTopic were highly correlated. This suggests that ChatGPT4Turbo interpreted the outputs of both models in a generally consistent way, even though the two models use different methods. Moreover, t-SNE visualisation (a method that represents high dimensional data in a two dimensional graph) showed that BERTopic produced more compact and clearly separated clusters than LDA, suggesting that BERTopic produces more coherent and distinct topics.

However, a later manual review showed that LDA was slightly more accurate than BERTopic, with topic relevance agreement of 89.8% compared with 84.9%. The authors then reveal that this difference may be related due to the use of BERTopic's default hyperparameters. They did not optimise parameters such as UMAP's n_neighbors or HDBSCAN's minimum cluster size, which may have influenced the topics produced by BERTopic. The authors also found that, unlike LDA, BERTopic did not require text preprocessing because its transformer embeddings could preserve the original sentence structure.

However, when applying BERTopic, hyperparameter tuning is important to identify the most suitable configuration for a given dataset. Different hyperparameter settings can produce different topic structures and outcomes; therefore, multiple configurations should be tested and compared to determine which results are the most coherent, meaningful, and interpretable. Consequently, relying on BERTopic's default hyperparameters may limit the quality of the resulting topics, making systematic tuning an important step in selecting the final model configuration.

3. Methodology

3.1 Research Approach

This project used BERTopic to identify meaningful themes across a large corpus of Creativity Support Tool (CST) evaluation literature, and whether those themes were also identified in Cox et al's (2025) manual thematic analysis.

The methodology covers five stages: data collection, text pre-processing, embedding model development, topic model implementation, and comparative evaluation against Cox et al. (2025).

3.2 Data Collection

Cox et al. (2025) identified 173 papers in their survey. However, upon reviewing the list of included papers in Table 1 in the survey paper, I identified a duplicate entry: paper 81 was listed twice. After removing this duplicate, the dataset contained 172 unique papers in total.

Abstract retrieval:

The DOIs for all 172 papers were compiled from the survey paper's reference list into a structured CSV file, alongside their reference numbers and publication years. Before proceeding with full text retrieval, abstracts were retrieved using OpenAlex, an open scholarly metadata database and API, as an intermediate validation step to confirm that the papers could be successfully accessed through the available APIs.As full text retrieval was expected to be more time consuming, this preliminary step allowed the accessibility of the papers to be assessed before initiating the full text retrieval process. Open Alex was able to retrieve all 172 papers with its bibliographic metadata, including title, abstract, publication year, and open-access status.

Full-text retrieval:

Full text was then retrieved for as many of the 172 papers as possible using a multi-stage pipeline. Automated API retrieval was first carried out using OpenAlex, and arXiv, an open-access repository of research papers. While OpenAlex flagged 78 papers as 'is_oa == False' meaning it could not find any version of the papers, Unpaywall, an open-access service that identifies legally available full-text versions of research papers, was accessed through Zotero and was able to locate open copies for these papers.

Full text was successfully obtained for 169 of the 172 papers (98.3%). The remaining three papers could not be retrieved as I was not able to find any accessible full text version through the sources used and were therefore excluded from the final corpus.

2.3 Text Pre-processing:

Prior to embedding and topic modelling, the 169 retrieved full texts were cleaned using Python's spaCy library to remove noise that would otherwise degrade clustering quality. Cleaning was implemented as a configurable pipeline of six toggleable stages:

- Reference removal: text from the first occurrence of a standard section heading (‘References', 'Bibliography ', 'Works Cited’, 'Literature Cited ') onward was discarded, removing citation lists in their entirety.
- Header and footer removal: publisher boilerplate was removed, including copyright notices, ACM permissions, ISBN lines and DOI URLs.
- Table removal: figure and table captions were removed, along with lines in which the majority of characters were numeric or symbolic rather than alphabetic.
- PDF artefact correction: hyphenation breaks introduced by line wrapping, ligature characters specific to PDF fonts, residual email addresses and URLs, and excess whitespace were removed. This stage also removed front matter such as author names and institutional affiliations appearing between the title and abstract.
- Normalisation: all text was lowercased.
- Tokenisation and lemmatisation: text was tokenised using spaCy's small English model (en_core_web_sm), with stopwords removed, followed by removal of punctuation and non alphabetic tokens, and lemmatisation of all remaining words to their dictionary root form.

This process substantially reduced document length: raw full text averaged 11,794 words per paper, while fully processed text averaged 4,751 words per paper.

Two distinct outputs were produced from this pipeline, reflecting the different requirements of BERTopic's embedding and keyword extraction components:

An embedding text version was produced by applying only stages 1 - 4 (structural cleaning: references, headers/footers, tables, and PDF artefacts) while explicitly skipping normalisation and lemmatisation, preserving natural sentence structure, casing, and grammar.

This was necessary because sentence transformer models are designed to generate embeddings from natural language, where contextual relationships and word order contribute to the representation of meaning. Applying aggressive lemmatisation and stopword removal could therefore alter the semantic information available to the embedding model.

A clean text version was also produced after applying the cleaning stages to implement a ‘bag-of-words’ style representation with count vectoriser using unigrams and bigrams. This allowed the model to identify meaningful individual single words and meaningful pairs of consecutive words to provide more specific information about the topics.

2.4 Embedding Model Development:

Document embeddings were generated using SentenceTransformers, a library that provides pretrained transformer models for converting text into numerical representations. An initial embedding pass used all-mpnet-base-v2, a widely used sentence transformer model producing 768 dimensional vectors. This model was selected initially because it is known and recommended to be the best embedding model for use with BERTopic. Its performance was also supported by research papers like Badlieah et al. (2024) which studied different embedding models and found that all-mpnet-base-v2 achieved the

highest performance in their study with a precision of 99.17%. However they used titles and abstracts only, whereas my study applied the model to full text papers. This revealed an important limitation of the model: it has a maximum input length of 384 tokens.
Given that the papers in this corpus averaged several thousand words after cleaning, most of the text in each paper was truncated before the embedding was generated. Therefore the resulting clustering of BERTopic was only representing the beginning of the text rather than the full text.

This led to changing the embedding model to BAAI/bge-m3, which supports an 8,192 token context window (approximately 21 times larger) and produces 1024 dimensional vectors. However, as most papers in the corpus still exceeded this limit, a chunking strategy was implemented: each paper's embedding text was divided into overlapping segments of approximately 2,000 words, with a 150 word overlap between consecutive chunks to reduce the loss of context.

Each chunk was embedded independently, and a paper's resulting chunk level embeddings were mean pooled into a single 1024 dimensional vector representing the paper as a whole, papers short enough to fit within a single chunk were unaffected by this process, as pooling a single vector simply returns that vector.

Overall, the chunking process produced 901 chunks from 169 papers, with an average of 5.3 chunks per paper. Only one paper fit within a single chunk while the longest required twelve, this approach allowed data from across all papers to be included in the embeddings while considering the embedding model's input length limit.

2.5 BERTopic model Implementation:

BERTopic was used to identify latent topics within the corpus. It consists of dimensionality reduction, clustering and topic representation.

The document embedding results were too high dimensional (1024 number per paper), therefore it makes it difficult for the clustering algorithm to work. This is because in high dimensional space, the distance between points becomes less significant. To address this, UMAP (Uniform Manifold Approximation and Projection) was implemented to reduce the dimensionality into a lower dimensional space while still maintaining the relative similarity between points (papers that were close together in the original dimensional space still remains close after the reduction). Using UMAP, the papers were each embedding to 5 dimensions using each paper's 15 nearest neighbours, a minimum distance of 0.0 to allow similar papers to be placed as close together as possible, and cosine similarity as the distance metric.

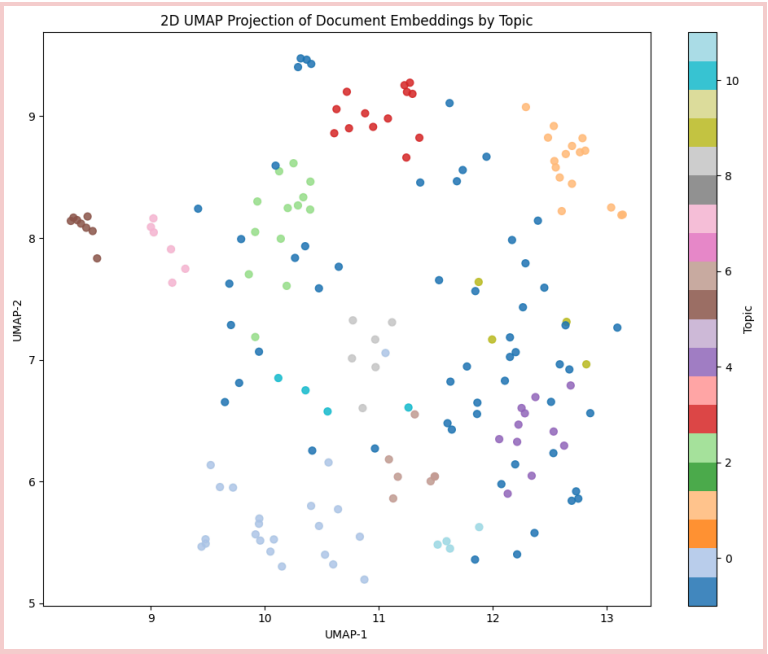


Figure 1: Two-dimensional UMAP projection of the original 1,024 dimensional document embeddings. Each dot is a document and the colours indicate the topic assignments produced by BERTopic.

These reduced embeddings were then clustered using HDBSCAN,which is a density based clustering algorithm that groups points together from denser regions of the data. This algorithm automatically groups them without someone requiring to specify a number for the clusters.

HDBSCAN was configured using euclidean distance to measure the similarity between papers and the excess-of-mass (EOM) method to select the most stable clusters across different density levels. This approach improves the reliability of the clustering results and reduces the effect of noise. HDBSCAN does not force every paper into a cluster, the papers that do not belong to a dense enough or well-defined group are categorised as outliers and assigned to Topic −1.

After the clusters were formed, c-TF-IDF, a version of the TF-IDF algorithm that evaluates the importance of the words in the clusters, was used to to extract topic keywords.These keywords help describe the main characteristics of each topic by identifying words that are more specific to one cluster than to the rest of the corpus. The same count vectorizer was used to generate the word representations for this step.

Since BERTopic does not require the exact number of topics to be specified in advance, the main hyperparameter that needed to be determined was HDBSCAN's minimum cluster size. This parameter defines the minimum number of papers required for a group to be identified as a cluster. The parameter was therefore tested at different values to determine which setting produced the most suitable clustering results.

Using the BERTopic default minimum cluster size of 5 produced eight topics, but 52 of the 169 papers (30.8%) were assigned to the outlier group (Topic −1). The minimum cluster size was therefore reduced to 4, which improved corpus coverage by reducing the number of outliers to 32 of 169 papers (18.9%) and resulted in the identification of one additional small topic. Based on these results, a minimum cluster size of 4 was selected for the final model.

Each topic the model produced was manually validated afterwards. Abstracts (and full texts, where needed) were reviewed against each topic's keywords and soft-clustering membership probabilities. This process enabled the initial topic interpretations to be confirmed, refined, or, where appropriate, challenged when the keywords did not capture the central theme of the paper. This manual verification process, along with the insights it provides about the strengths and limits of automated topic modelling in this domain, forms an essential component of the analysis and discussion presented in this thesis.

4. BERTopic results

4.1 Thematic Review:

To contextualise the topic model against existing literature, its output was compared with Cox et al's (2025) systematic review of outcome measures used across 173 Creativity Support Tool (CST) evaluations published in the ACM Digital Library between 2015 and 2024. Of the 173 papers included in that review, one was a duplicate that was not corrected in the published paper, leaving 172 unique papers; 169 of these were retrievable and used as the training corpus for BERTopic modelling.

4.2 Topic Model Output

BERTopic identified ten substantive topics (Topics 0–9, n=137) plus a residual outlier cluster (Topic -1, n=32) that the algorithm could not confidently assign to any topic. Topic sizes ranged from 42 papers (Topic 0, design ideation tools) down to 4 papers (Topic 9, animation and character design/motion).

Table 1 summarises each topic's label, top keywords, size, and the years in which it is concentrated:

Topic	Label (manual)	Top keywords	Num.papers	Dominant years
0	Design Ideation Tools: Generative AI and Non-AI Approaches	['board', 'analogy', 'mood board', 'mood', 'generative ai', 'fig', 'design space', 'generate image', 'adjective', 'conceptual']	42	2016–24
1	Immersive environments and Musical CSTs	['vr', 'sound', 'musical', 'visualization', 'adaptive', 'spatial', 'rice', 'factor', 'ar', 've']	22	2016–2024
2	Stimulus Driven Ideation Support	['crowd', 'ideator', 'idea generation', 'partner', 'worker', 'generate idea', 'chat', 'difference', 'facilitator', 'mind']	16	2015–2023
3	CSTS for creative writing	['character', 'story', 'viewer', 'narrative', 'chart', 'writer', 'streamer', 'stream', 'live', 'news']	13	2015–24
4	LLM based writing for broad topics	['writing', 'writer', 'metaphor', 'journalist', 'story', 'variation', 'llm', 'inject', 'spark', 'language model']	13	2015–24
5	Social Robots for children's creativity	['robot', 'child', 'social robot', 'drawing', 'non creative', 'creative condition', 'jibo', 'child creativity', 'measure', 'story']	8	2019–24

6	CSTs for color/painting decision making	['palette', 'painting', 'paint', 'colorization', 'symbol', 'brush', 'drawing', 'gradient', 'novice', 'chinese']	7	2017–23
7	Software-Based Creativity Support Tools for Children	['child', 'story', 'avatar', 'scratch', 'storytelling', 'collage', 'programming', 'character', 'drawing', 'visual storytelling']	6	2017–24
8	The Human-AI Relationship in Co-Creative Collaboration	['co creative', 'human ai', 'agent', 'creative system', 'drawing', 'robot', 'ai agent', 'wizard', 'reframer', 'initiative']	6	2018-2024
9	CSTs for Animation/Character Design and Motion t	['animation', 'puppet', 'rig', 'character', 'template', 'shadow', 'pose', 'shadow puppet', 'cinematic', 'secondary']	4	2015–24
−1	Outliers (unclustered)	llm, edit, template, ml, viewer, animation, palette, drawing, story, risk	32	2015–24

Table 1. BERTopic cluster overview (n=169 corpus, 10 topics + outlier cluster).

Figure 2 and 3 show the distribution of the topics and important key words identified my BERTopic. Almost one fifth of the corpus (32/169 papers, ≈18%) fell into the unclustered outlier group. Thus, roughly four fifths of the corpus can be interpreted with confidence in relation to the survey's claims, while the remaining fifth is not represented by any coherent topic and is therefore excluded from the comparison.

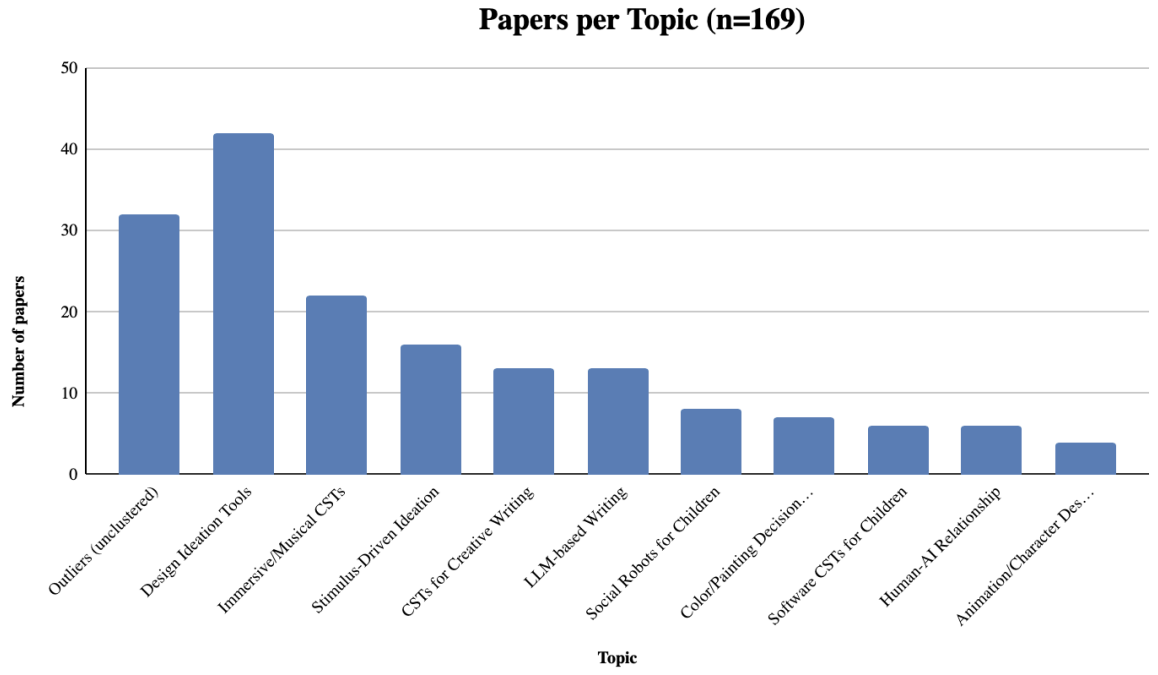


Figure 2 . Topics size distribution

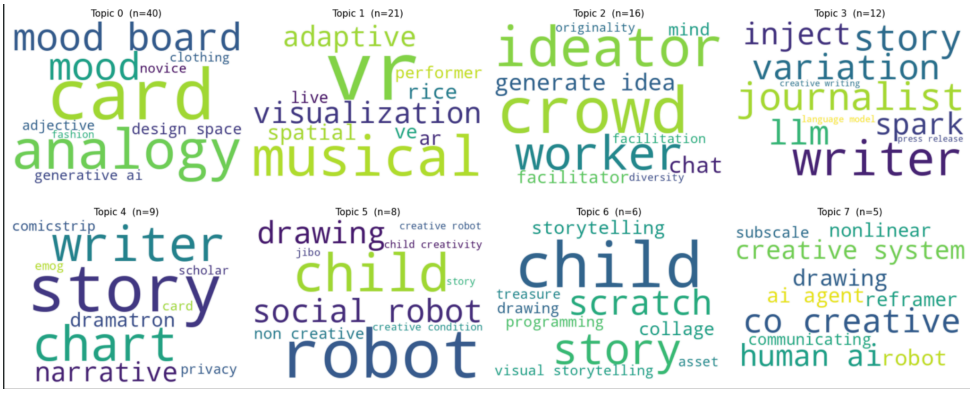


Figure 3: Word clouds representing the most important key terms according to BERTopic

4.3 Analysis and Interpretation of BERTopic Results

Interpretation of the topic clusters: Manual verification and Name labelling decision

In this section, I unfold the results of my BERTopic model and show how these were interpreted and labeled manually. From the papers in each cluster, I've chosen the papers with the highest probability rate from the soft clustering. Meaning, my model doesn't only give a yes or no label of whether it belongs to a certain cluster but it also finds a degree of membership between 0 and 1 which represents how strongly it represents that cluster.
For example, a probability of 1 represents that the paper sits in the core zone of the cluster and a lower probability means it is more peripheral and sits further away. Depending on how close a paper is to the boundary between clusters, it may potentially be assigned to a different cluster. Therefore, this section focuses on the papers within each cluster that are closest to its core, in order to assess whether BERTopic has assigned them appropriately.

The following tables present papers with a membership probability of 1 where sufficient examples were available. For clusters with fewer papers having a probability of 1, the remaining examples were selected depending on their proximity to the cluster core, using the next highest membership probabilities where necessary. Not all papers within each cluster are shown, as some clusters contain a large number of papers, the tables therefore provide a representative selection of papers closest to the core rather than the complete membership of each cluster.

Topic 0: Design Ideation Tools: Generative AI and Non-AI Approaches

Key words: 'board', 'analogy', 'mood board', 'mood', 'generative ai', 'fig', 'design space', 'generate image', 'adjective', 'conceptual'

Num ref	paper	year	Topic	Prob.	ideation and design related	gen AI related
31	AutoSpark: Supporting Automobile Appearance Design	2024	0	1.0	Yes	Yes

	Ideation with Kansei Engineering and Generative AI					
33	Beyond Numbers: Creating Analogies to Enhance Data Comprehension and Communication with Generative AI	2024	0	1.0	Yes	Yes
36	CreativeConnect: Supporting Reference Recombination for Graphic Design Ideation with Generative AI	2024	0	1.0	Yes	Yes
63	How Digital Tools Impact Convergent and Divergent Thinking in Design Ideation	2021	0	1.0	Yes	No
98	f3.js (Web-based design tool called f3.js)	2017	0	1.0	Yes	No
117	3DALL-E: Integrating Text-to-Image AI in 3D Design Workflows	2023	0	1.0	Yes	Yes
118	Design Space Cards	2021	0	1.0	Yes	No

Table 2 . Topic 0 - overview of the papers with the highest degree of cluster membership

Topic 0 contains the most papers compared to the rest of the clusters, 42 papers about CST. Just looking at the paper titles and after reading the abstracts, we get an idea that the papers might be about using AI or a different tool for the purpose of designing. After a manual review, I found that while every paper in this cluster relates to design ideation, not all of them involve generative AI. A group of papers apply generative AI to the ideation process (e.g., paper 31, '*AutoSpark'*, and paper 117, '*3DALL-E'*), while the remaining papers explore the support for ideation process through other technical and physical tools (e.g., paper 98 '*f3.js*' and paper 118 '*Design Space Cards'*) like using deck of cards to brainstorm, random sampling, platforms like 'lumies' for game design and technical tools like 'f3.js' which is a web based design tool.

Therefore, the two sub-themes emerged from the topic analysis:

1. Gen AI supported design ideation: where generative AI models produce content, suggestions, or inspiration within the ideation process.
2. Non-AI tool mediated design ideation: where the focus is on digital and software tools such as web based design tools and software gaming tools.

Topic 1: Immersive environments and Musical CSTs

BERTopic's HDBSCAN clustering assigned 22 papers to topic 1, with the C-TF-IDF top keywords being 'vr', 'sound', 'musical' and 'visualization'. At first glance, I could easily name the topic 'VR used as a CST in the music and media industry.

However, after conducting a thorough manual review of the abstracts and cross checking HDBSCAN's soft clustering probability outcomes, I found that a small number of papers sit right at the intersection between both themes and apply VR/AR specifically to musical composition and production, (e.g., paper 159, *Flow with the Beat!*, and paper 158, *XR Composition in the Wild*).

Two sub-themes emerged from the topic analysis:

1. Immersive environments for creativity support: Where VR is used to support ideation of creativity.
2. Musical and sound creativity support: where the focus is on generative tools for musical composition.

Key words: ‘'vr', 'sound', 'musical', 'visualization', 'adaptive', 'spatial', 'rice', 'factor', 'ar', 've'

Num ref	paper	year	Topic	Prob.	Vr related	Music related
21	Stimulating Creative Partnerships in Human-Agent Musical Interaction	2016	1	1.0	NO	YES
60	Reflection Across AI based Music Composition	2024	1	1.0	NO	YES
94	Multiplexed VR: Individualized Multiplexing Virtual Environment to Facilitate Switches for Group Ideation Creativity	2024	1	1.0	YES	NO
159	Flow with the Beat! Human-centered Design of Virtual Environments for Musical Creativity Support in VR	2022	1	1.0	YES	YES
185	SelVReflect: A Guided VR Experience Fostering Reflection on Personal Challenges	2023	1	1.0	YES	NO
207	Supporting Non-musicians? Creative Engagement with Musical Interfaces	2017	1	0.710845874948140 0	NO	YES
158	XR Composition in the Wild: The Impact of User Environments on Creativity, UX and Flow during Music Production in Augmented Reality	2024	1	0.5634964578898110	YES	YES

Table 3 . Topic 1 - overview of the papers with the highest degree of cluster membership

The remaining papers, by contrast, align with one sub-theme or the other, but not both, as shown in Table 3.

Reflecting on this pattern, it is possible that the shared HCI vocabulary used across these papers might've caused the embedding algorithm to cluster these sub-themes together in one topic.

Topic 2: Stimulus Driven Ideation Support

Key words: 'crowd', 'ideator', 'idea generation', 'partner', 'worker', 'generate idea', 'chat', 'difference', 'facilitator', 'mind'.

Num ref	paper	year	Topic	Prob.	Stimulus related	Ideation related
30	SearchIdea: An Idea	2023	2	1	Yes	Yes

	Generation Tool to Support Creativity in Academic Search					
69	The Effect of Peripheral Micro-tasks on Crowd Ideation	2017	2	1	Yes	Yes
70	CrowdMuse	2019	2	1	Yes	Yes
170	Providing Timely Examples Improves the Quantity and Quality of Generated Ideas	2015	2	1	Yes	Yes

Table 4 . Topic 2 - overview of the papers with the highest degree of cluster membership

While the key words for this topic ‘ideator’ and ‘idea generation’ might be similar to topic 0, there is an underlying reason why these papers have been clustered together which BERTopic might’ve not been able to understand. Unlike the other topics, this one is a lot more subtle as the title of the papers does not give it away as easily. However, all these papers are studying how a stimulus of some kind (e.g., crowd idea combination, inspiration on past, or timing of delivery etc) can affect the output of creativity.

In a nutshell, BERTopic's clustering is consistent with the Stimulus - Outcome pattern I identified across these papers, but this is likely because of the similarity in language the papers used to describe idea generation and stimuli and not because the algorithm understood the underlying pattern or context. Therefore, the clustering is a byproduct of lexical/semantic similarity in how these ideas are described, rather than evidence that the model recognized the actual research design similarity.

Topic 3: CSTS for creative writing

Key words: 'character', 'story', 'viewer', 'narrative', 'chart', 'writer', 'streamer', 'stream', 'live', 'news'

Num ref	paper	year	Topic	Prob.	Writing	AI
133	Co-Writing Screenplays and Theatre Scripts with Language Models: Evaluation by Industry Professionals	2023	3	1	Yes	Yes
152	CharacterMeet: Supporting Creative Writers' Entire Story Character Construction Processes Through Conversation with LLM powered Chatbot Avatars	2024	3	1	Yes	Yes
189	ReelFramer: HumanAI Co Creation for News to Video Translation	2024	3	1	Yes	Yes
197	wr-AIter: Enhancing Ownership Perception in AI-Driven Script Writing	2024	3	1	Yes	Yes
83	Portrayal: Leveraging NLP and Visualization for Analyzing Fictional Characters	2023	3	1	Yes	No

Table 5 . Topic 3 - overview of the papers with the highest degree of cluster membership

Topic 3's keywords appear to involve some form of creative writing; however, there is no sign of 'AI' or 'LLM,' which raises a flag after reading the titles of the papers with a soft clustering probability of 1 as 4 out of 5 of those definitely use generative AI to cowrite stories or scripts. This is likely because

C-TF-IDF highlights words that are distinctive to a cluster compared to others, rather than words that simply appear frequently throughout the text.

As AI related words like 'AI generation', 'idea generation', 'LLM', or 'AI agent' already show up in other clusters, as seen in previous topics, my assumption is that they aren't distinctive enough to surface as keywords for Topic 3, even though, to human interpretation, this is clearly a notable and central theme.

After a manual review of the abstracts, I found that while the papers all relate to writing and character development, the way AI is implemented differs. The first four papers in the table use generative AI for narrative and story development, and for generating dialogue for characters alongside the human writer. The fifth paper, *Portrayal: Leveraging NLP and Visualization for Analyzing Fictional Characters*, uses natural language processing analytics to interpret existing content rather than generate new content.

Although the majority of the papers used AI either directly, as a supporting tool, or to interpret content, I named the topic CSTs for Creative Writing, since ultimately all five function as support tools for the creative writing process.

Topic 4: LLM based writing

Key words: 'writing', 'writer', 'metaphor', 'journalist', 'story', 'variation', 'llm', 'inject', 'spark', 'language model'

Num ref	paper	year	Topic	Prob.	AI/LLM	Writing tool
25	Creativity Support in the Age of Large Language Models: An Empirical Study Involving Professional Writers	2024	4	1.0	yes	yes
67	Sparks: Inspiration for Science Writing using Language Models	2022	4	1.0	Yes	Yes
155	ABScribe: Rapid Exploration & Organization of Multiple Writing Variations in Human AI Cowriting Tasks using Large Language Models	2024	4	1.0	Yes	Yes
226	VISAR: A Human-AI Argumentative Writing Assistant with Visual Programming and Rapid Draft Prototyping	2023	4	1.0	Yes	Yes
73	You're the Voice	2015	4	0.523708386 6046900	No	Yes

Table 6 . Topic 4 - overview of the papers with the highest degree of cluster membership

The majority of papers from Topic 4 are about using LLMs to assist writers in brainstorming, drafting, or inspiring science papers. This might seem like an overlap with Topic 3, which involves CSTs for creative writing. However, while Topic 3 focuses on narrative, storyline building, and character

development, Topic 4 is more generic, ranging from scientific writing to journalistic writing. When my model clustered Topic 3 and Topic 4 separately, I was initially confused, expecting them to merge into one. However, even after tuning my hyperparameters multiple times and changing the minimum cluster size in the embedding model, it consistently produced the same outcome.

Topic 5: Social Robots for children's creativity

Key words: 'robot', 'child', 'social robot', 'drawing', 'non creative', 'creative condition', 'jibo', 'child creativity', 'measure', 'story'

Num ref	paper	year	Topic	Prob.	Children	Bots
2	Escape!Bot: Social Robots as Creative Problem-Solving Partners	2022	5	1.0	Yes	Yes
3	Can Children Learn Creativity from a Social Robot?	2019	5	1.0	Yes	Yes
5	Creativity Encounters Between Children and Robots	2020	5	1.0	Yes	yes
53	CreativeBot: a Creative Storyteller robot to stimulate creativity in children	2022	5	1.0	Yes	yes
84	Exploring the Role of Social Robot Behaviors in a Creative Activity	2021	5	1.0	Yes	yes

Table 7 . Topic 5 - overview of the papers with the highest degree of cluster membership

This topic, with only 8 papers, focuses on the study of how social robots interact with and support children's creativity. The summary table only shows the papers situated at the core of the cluster, with a probability of 1; however, after manual evaluation, I confirmed that all 8 papers share the same themes: children as the users, and AI robots supporting children's creativity.

Unlike topic 3, where naive keyword readings diverged from the papers' actual content, Topic 5 is a case where BERTopic's output and manual review align closely, and the keywords ('child', 'social robot', 'child creativity') reflect the shared focus across all papers in the cluster. However, it's worth noting the keyword 'Jibo', a commercial robot, which seems to only appear in paper 2, '*Escape!Bot'* , yet is apparently relevant enough by BERTopic's judgement to be included among the topic's most important and frequent keywords.

Topic 6: CSTs for color/painting decision making

Key words: 'palette', 'painting', 'paint', 'colorization', 'symbol', 'brush', 'drawing', 'gradient', 'novice', 'chinese'

Num ref	paper	year	Topic	Prob.	Painting
64	Painting an Apple with an Apple	2018	6	1.0	Yes
76	Creative and Progressive Interior Color Design with Eye-tracked User Preference	2022	6	1.0	Yes
80	CoColor: Interactive Exploration of Color Designs	2023	6	1.0	Yes
166	Playful palette	2017	6	1.0	Yes

167	Color Builder	2019	6	1.0	Yes

Table 8 . Topic 6 - overview of the papers with the highest degree of cluster membership

Topic 6 focuses on interactive creative support tools that help decision making when painting and designing, and the top keywords reflect this. However, the keywords 'symbol' and 'chinese' seem out of place and are worth flagging for a manual check.

Interestingly, these words are most prominent in Magical Brush, one of the two lowest probability papers in the cluster (0.2) and notably, it is also the only paper that breaks from the cluster's otherwise consistent non AI, interaction driven theme, since it combines generative AI with symbolic Chinese painting elements. The other low probability paper, CogCues, remains consistent with the cluster's core theme despite also sitting at the periphery.

This suggests that despite sitting at the edge of the cluster, C-TF-IDF can allow a single distinctive paper (Magical Brush) to disproportionately shape the topic's keywords. This is the contrary effect to Topic 3's pattern, where a theme shared across most papers (AI use) failed to surface as a keyword because it was too common across the corpus to be distinctive.

Topic 7: Software Based Creativity Support Tools for Children

Key words: 'child', 'story', 'avatar', 'scratch', 'storytelling', 'collage', 'programming', 'character', 'drawing', 'visual storytelling'

Num ref	paper	year	Topic	Prob.	Children
32	ChatScratch: An AI-Augmented System Toward Autonomous Visual Programming Learning for Children Aged 6-12	2024	7	1.0	Yes
65	Create to learn Paradigm: A Proxy Visual Storytelling Tool (PVST) for Stimulating Children's Story Sense and Structure	2024	7	1.0	yes
138	Augmented Games	2017	7	1.0	Yes
222	StoryDrawer	2022	7	1.0	yes
220	Investigating the Effects of Self Avatars and Story Relevant Avatars on Children's Creative Storytelling	2020	7	0.6117194010440770	yes

Table 9 . Topic 7 - overview of the papers with the highest degree of cluster membership

The keywords of this topic are accurate and reflect the papers well. However, it's unusual that there are no AI related keywords, even though two papers are centrally related to generative AI. This might be the same reasoning as in Topic 3, where AI related words weren't distinctive enough to surface as keywords.

Similar to Topic 5, all papers in this topic involve children as the study population, but the main difference is that they aren't using physical robots, instead, software based tools like storytelling interfaces, programming platforms, and drawing systems are used. Since the underlying technology

differs, the papers are classified into different topics. Although the exact reason for this distinction is unclear, one possible explanation is that BERTopic's embeddings distinguish between clusters primarily depending on the domain or type of technology being used, rather than on the study population, even when the population is the same across both topics.

Topic 8: The Human AI Relationship in Co-Creative Collaboration

Key words: 'co creative', 'human ai', 'agent', 'creative system', 'drawing', 'robot', 'ai agent', 'wizard', 'reframer', 'initiative'

Num ref	paper	year	Topic	Prob.	Human-ai relation ship
114	It Is Your Turn: Collaborative Ideation With a Co-Creative Robot through Sketch	2020	8	1.0	yes
139	I Lead, You Help but Only with Enough Details	2018	8	1.0	yes
156	Understanding User Perceptions, Collaborative Experience and User Engagement in Different Human-AI Interaction Designs for Co-Creative Systems	2022	8	1.0	yes
229	Understanding Nonlinear Collaboration between Human and AI Agents: A Co-design Framework for Creative Design	2024	8	1.0	Yes

Table 10 . Topic 8 - overview of the papers with the highest degree of cluster membership

Looking deeply into the papers in this topic, unlike previous topics, it doesn't specifically involve one technology or one subject. Instead, the common thread throughout the papers is studying the mechanics of the collaboration process itself, meaning the 'product' in these papers is almost secondary; the actual research question is how humans and AI should interact as co-creative partners, independent of domain.

This is an interesting finding on how the model works: unlike Topic 5 and Topic 6, where cluster coherence was driven by a common subject (children, color), Topic 8's coherence is driven by a shared research question seen across all papers in the cluster, even though the subject of each paper differs. Despite this, BERTopic still grouped them together accurately. This suggests that, in this instance, there is a possibility that the embedding model captured a genuine conceptual similarity in research framing, rather than relying solely on domain specific vocabulary.

Topic 9: CSTs for Animation/Character Design and Motion

Key words: 'animation', 'puppet', 'rig', 'character', 'template', 'shadow', 'pose', 'shadow puppet', 'cinematic', 'secondary'

Num ref	paper	year	Topic	Prob.	Animation design
79	An Interactive System for Supporting Creative Exploration of Cinematic Composition Designs	2024	9	0.6551854611941540	yes
99	TextAlive	2015	9	0.6551854611941540	yes

200	Secondary Motion for Performed 2D Animation	2017	9	0.6551854611941540	yes
217	Lumina: A Software Tool for Fostering Creativity in Designing Chinese Shadow Puppets	2024	9	0.6551854611941540	yes

Table 11 . Topic 9 - overview of the papers with the highest degree of cluster membership

This topic, as seen from the table, did not have any paper with a cluster membership probability of 1; instead, all four papers were split equally with a probability of 0.65 each. This is unusual, and it means they sit at the exact same position and distance from the cluster's core structure. My assumption for this result is that it perhaps happened because the cluster is too small.

Overall, manually checking the results was key to actually understanding the output BERTopic. A lot of the time, the keywords grouped different themes together or missed an important detail, especially AI use, which showed up in so many papers across the whole dataset that it stopped standing out as a ‘distinctive’ keyword. On the flip side, sometimes a single paper with unusual wording ended up shaping a topic's keywords way more than it should have, even when that paper barely fit the cluster.

That said, BERTopic wasn’t always wrong, in a few cases the keywords matched the papers well, and it even picked up on papers that shared a similar research approach despite covering totally different subjects. So overall, BERTopic was a good starting point, but manual review was what actually made sense and use of it.

Topic -1: Outliers

Key words: 'llm', 'ml', 'comic', 'risk', 'template', 'edit', 'child', 'story', 'toolkit', 'chatgpt'

Num ref	paper	year	Topic	Pro.	Accessibility and disability focused CST	LLM assisted co creativity
121	Physical, digital, and hybrid setups supporting card-based collaborative design ideation	2018	-1	0.3295477813904880		YES
134	Empowering Expression for Users with Aphasia through Constrained Creativity	2019	-1	0.07258665116625380	YES	
135	CreaTable Content and Tangible Interaction in Aphasia	2020	-1	0.06295075610190270	YES	
179	Accessible Creativity with a Comic Spin	2020	-1	0.07032368501796210	YES	
180	Comic Spin: A Comic Creation Tool Enabling Self-expression for People with Aphasia	2022	-1	0.07941808377627030	YES	
213	Jamplate: Exploring LLM-Enhanced Templates for Idea Reflection	2024	-1	0.06382738994874510		YES

Table 12 . Topic -1 (Outliers) : sample overview of the papers

The outlier papers have very low membership probabilities, with paper 121 having the highest at 0.33. This means none of them fit confidently into any cluster. Meanwhile, smaller topics like Topics 5 to 9 (all under 8 papers) have the most confident probabilities compared to larger topics like 0 and 1, which have far more spread out scores. This suggests small clusters tend to be more confident and uniform. Topic 9 is an example of this, where all four papers landed on the exact same probability score of 0.66, treating all the papers uniformly.

The outlier papers, on the other hand, do show real themes, just ones too small to form their own clusters. For example, there's a small group of papers on accessibility and disability focused tools (including work on aphasia and comic based self expression) that BERTopic didn't detect as its own theme, and a group on LLM assisted co creativity (including ‘Luminate’ and ‘Jamplate’) that sits somewhere between Topics 0 and 4 but wasn't picked up either.

4.4 Comparative Analysis: Cox et al.’s Thematic Analysis and BERTopic

Out of the 169 papers the model was trained on, BERTopic classified 137 of those into clusters, leaving 32 papers as anomalies that could not be assigned to any cluster.

The main distinction between the two approaches is the question each addresses. Cox et al.'s three main themes: user experience, artefact quality, and user centric benefits, categorises papers by how success was measured. My model, by contrast, categorises papers by the lexical content of the full text, which in this corpus predominantly reflects the subject of the paper, such as the type of technology or domain they address. Although the two approaches address different questions, there are still several notable areas of overlap that could complement and support one another.

One of the more obvious overlaps is the large number of studies on children's creativity. In Cox et al.'s analysis, they found that user centric benefit measures (such as self reflection, well being, and intrinsic ability) were particularly prevalent in studies where children were the user group. Quite similarly, BERTopic clustered children focused papers into two separate topics : Topic 5 (social robots for children's creativity) and Topic 7 (software based creativity tools for children), without any knowledge of Cox et al.'s findings. This suggests that children's creativity is a recurring and distinct pattern within the CST corpus.

Another overlap worth mentioning is Cox et al.'s discussion of human AI co-creativity. In the survey paper, they mention AI as a creative partner rather than a tool, citing paper 156, *‘Understanding User Perceptions’* by Rezwana and Maher, as an example. This same paper appears in Topic 8 of my model, with a soft clustering membership probability of 1, meaning it sits close to the core of the cluster. This suggests that the definitional distinction raised by Cox et al. was also detectable by BERTopic.

A third overlap I found relates to a similar underlying issue Cox et al. identified, the difficulty of consistently capturing AI's role within CST evaluations. They discuss how the increasing use of generative AI in creativity has made CST evaluation harder, so many researchers improvise, relying on ad hoc, one off measures to capture concerns related to AI collaboration. For example, rather than using a standardised, validated measure of human AI co-creativity, researchers often ask participants to rate agreement with statements like ‘I felt like I was collaborating with the AI system’ (Mirawsky et al.). This statement from paper 133 suggests that it becomes harder to compare results across studies when there isn't a standardised set of questions being used to measure the same underlying concept.

A similar pattern emerged throughout my analysis of the topics, as several topics contained a mixture of AI and non AI papers clustered together in one topic, likely because they shared the same domain. For example, Topic 0 grouped AI driven tools like ‘AutoSpark’ alongside non AI tools like the

‘Design Space Cards deck’ as they both shared Ideation Design CSTs as their domain, and Topic 6 grouped an AI assisted painting system (Magical Brush) alongside otherwise non AI colour tools. In two cases, this instability was more pronounced, for example: in Topic 3, four of five papers relied centrally on generative AI, yet no AI related term appeared among the topics defining keywords; the same pattern occurred in Topic 7, where two AI driven tools (ChatScratch and StoryDrawer) sat alongside non AI tools without AI ever surfacing as a distinctive keyword.

5. Findings and Discussion

The findings suggest that BERTopic can provide a useful initial representation of the structure of a research literature, but that the structure it produces depends strongly on what is present in the language of the papers. Across the ten substantive topics, BERTopic successfully separated the corpus into recognisable groups, with several clusters showing clear thematic coherence.

For example, Topic 5 brought together papers concerning social robots and children's creativity, while Topic 6 captured papers focused on colour and painting related creativity support.

In these cases, the lexical and semantic similarities identified by the model corresponded closely with the themes identified through manual review. However, other clusters revealed a more complicated relationship between the language of a paper and the underlying research question being investigated.

A key finding is therefore that the focus of a paper matters when determining what should count as a theme. BERTopic primarily groups documents according to similarities in their textual representations. As mentioned in the methodology, this means that papers may be clustered because they discuss a similar subject, application domain, or similar terminology, without necessarily sharing the same research purpose.

This explains why the BERTopic results cannot be expected to reproduce Cox et al.'s themes directly. A paper about children's creativity, for example, may be placed in a BERTopic cluster because it discusses children, robots, storytelling, or a creative tool. However, from Cox et al.'s perspective, the same paper may be relevant because of the particular way it evaluates user experience or user centric benefits. The subject of the paper and the focus of its evaluation are therefore not necessarily the same thing. This suggests that an unsupervised topic model is capable of identifying what papers tend to be about, but is less reliable at determining what researchers are trying to investigate about that subject.

Nevertheless, the results also demonstrate that this limitation should not be interpreted as meaning that BERTopic is incapable of capturing research level similarities. Topic 8 provides an important example. The papers in this cluster cover different subjects and technologies, but share an emphasis on the mechanics of human AI collaboration in creative processes. Here, the model appeared to capture a similarity in research framing rather than simply grouping papers according to a common application domain. Even though it remains inconclusive, the results could imply that BERTopic might sometimes recover a shared research focus when that focus is sufficiently reflected in the terminology used across the papers. The issue, therefore, is not that semantic research focus is entirely inaccessible to the model, but that it is not guaranteed to be the organising principle of the resulting clusters as the terminology might always be related to the research focus.

The comparison also demonstrates why the model's output requires interpretation rather than being treated as self explanatory. Topic 2 is an example. Its papers were found through manual review to share a stimulus outcome pattern: they investigate how different forms of stimulus, such as crowd generated ideas, inspiration, or the timing of information delivery, affect creative output. BERTopic produced a coherent cluster, but the analysis could not determine that the model had recognised this underlying research design. Instead, the clustering may have resulted from similarities in the language used to describe idea generation and stimuli. The distinction is important because a researcher interpreting the cluster might reasonably identify a deeper methodological or theoretical theme that is not explicitly represented in the model's topic label.

The topic keywords themselves further demonstrate the importance of human interpretation. In Topic 3, four of the five centrally located papers relied on generative AI, yet AI related terms did not appear among the defining keywords. Conversely, Topic 6 contained the keywords 'symbol' and 'Chinese', which appeared to be disproportionately influenced by a single peripheral paper. This shows that topic keywords should not automatically be interpreted as a complete description of a cluster. A keyword can be absent despite being conceptually important, or appear because of its distinctiveness in only a small part of the cluster.

The outlier group provides another indication of the difference between automated clustering and human interpretation. Thirty two of the 169 papers were classified as outliers rather than being assigned to one of the ten substantive topics. Manual inspection showed that these papers were not necessarily irrelevant or thematically meaningless. For example, a number of accessibility and disability focused CST papers formed a recognisable small group, while other outliers concerned LLM assisted co creativity. These patterns appear meaningful from a research perspective but were too small or insufficiently similar to form separate clusters under the model. This is particularly relevant when considering Cox et al.'s survey, because the survey categorised papers according to evaluation measures rather than topical similarity. A paper may therefore be highly relevant to one of Cox et al.'s analytical themes while remaining difficult for BERTopic to place within a sufficiently large topical cluster.

These findings suggest that BERTopic is best viewed as an initial exploratory stage of thematic analysis rather than a direct substitute for human interpretation. Its main strength lies in rapidly exposing patterns within a large corpus that might otherwise be difficult to see. It can reveal concentrations around particular technologies, user groups, application domains, and, in some cases, shared research approaches. However, interpreting why those papers belong together and deciding whether that similarity is actually relevant to the research question still requires human judgement. This was evident throughout the manual validation process, where reviewing abstracts and full texts allowed me to interpret the initial topic and decide whether it had to be refined.

This has an important implication for researchers using AI assisted thematic analysis. The appropriate use of BERTopic may depend on the analytical focus established before the modelling begins. If the aim is to understand the general topical structure of a field, the clusters generated by BERTopic can provide a useful overview of the corpus. If, however, the research question concerns a more specific analytical dimension, the BERTopic clusters alone may not provide an appropriate classification of the data.

In such cases, however, BERTopic can still serve a valuable methodological purpose: the clusters can be used to organise and reduce the corpus, allowing researchers to identify and retain the most

relevant documents for the analysis. This can substantially reduce the volume of material that requires closer qualitative examination, thereby saving time while ensuring that the final thematic analysis remains aligned with the research question.

One possible approach would be to use BERTopic to identify and organise groups of papers, and then use an LLM to interrogate the papers within each cluster according to a predefined analytical focus. For example, rather than asking an LLM simply to describe Topic 3, a researcher could provide the abstracts or full texts from that cluster and ask the model to identify evidence relating specifically to user experience, evaluation measures, or human AI collaboration. This would combine the scalability of automated clustering with a more targeted semantic interpretation. A similar approach has been explored by Ma et al. (2025), whose study integrated ChatGPT 4Turbo with BERTopic to assist in interpreting the topics generated from their corpus. The approach is therefore potentially useful as a model for future AI assisted thematic analysis, although its reliability would still depend on careful prompting and human validation.

The role of the human researcher consequently remains important. LLM interpretation introduces its own risks, including the possibility of hallucinating and generating unsupported interpretations. Any LLM assisted stage should therefore require researchers to verify claims against the source papers and request evidence or references for interpretations rather than accepting generated summaries at face value. The aim should not be to remove human thematic analysis, but to reduce the amount of manual work required before and during it.

6. Implications, Limitations and Future Work

The findings indicate several practical implications for the use of BERTopic in literature research. First, BERTopic could be used as a screening and exploratory tool before an in depth literature review. Rather than manually reading every paper and only subsequently discovering that some are outside the scope of the research question, researchers could use topic modelling to identify broad clusters and potentially prioritise papers for closer examination. This could make the early stages of a large scale review more efficient, particularly when the initial corpus contains hundreds or thousands of papers.

Second, BERTopic could form one stage within a broader human-AI thematic analysis workflow. Meaning, the model could first organise a corpus into groups, after which researchers could interrogate each topic individually using an LLM according to a clearly defined research focus. This would be particularly useful where the research question concerns a dimension that is not necessarily aligned with the topic structure of the literature. However, such a workflow should also retain human verification at the interpretation stage.

The study also has its limitations. Most importantly, the comparison with Cox et al. is not a direct comparison between two methods attempting to solve exactly the same analytical problem. Cox et al.'s themes are manually constructed around outcome measures, whereas BERTopic's clusters emerge from the textual similarity of the papers. Therefore, a lack of correspondence between a BERTopic topic and one of Cox et al.'s themes should not necessarily be interpreted as a failure of BERTopic. Rather, it demonstrates that different analytical methods can produce different representations of the same literature depending on the question they are designed to answer.

Another limitation is the presence of outliers that were not assigned to a substantive topic, meaning that a meaningful portion of the literature was not represented by the ten main clusters. Manual examination showed that some of these papers contained meaningful themes that were simply too small or heterogeneous to form their own clusters. Future studies could therefore investigate whether alternative clustering parameters, embedding strategies, or hierarchical approaches could recover these smaller research areas without reducing the coherence of the larger topics.

7. Conclusion

The development of this study answers the research questions by showing that BERTopic and the manual thematic analysis by Cox et al. (2025) identify overlapping patterns but organise the literature according to different structures, as predicted. The survey paper primarily organises the papers according to outcome measures relevant to the CSTs, whereas BERTopic appears to organise papers more strongly according to the application or technological domain.

Therefore, these findings reveal that BERTopic is a useful exploratory tool for analysing large corpora, but not fully autonomous, as human thematic analysis is still required to supervise the model. Its value lies in revealing patterns and reducing the initial burden of working through a large literature corpus. Human interpretation is still necessary, as these patterns cannot be interpreted from the clusters alone. A ‘human-in-the-loop’ therefore remains necessary to establish why papers have been grouped together, whether the resulting patterns address the research question, or whether they only reflect lexical and semantic similarities. To translate these semantic patterns into meaningful findings relevant to the research question, human interpretation is required to determine what theoretical significance should be assigned to them.

This is reflected in the data evaluation, where there were differences between the findings generated by the BERTopic model in this study and the manual thematic analysis conducted by Cox et al. (2025). Although there were overlaps between the topics identified by the model and the themes reported by Cox et al., these overlaps were not always directly apparent from the model outputs. Human interpretation was therefore required to connect the identified topics to the broader findings and to assess their relevance to the research question concerning CST evaluation.

This study supports the finding that BERTopic modelling clusters topics according to the subject or domain of the paper. This is clearly seen when it distinguishes and separates papers related to child users into two topics, although they have the same population, because the domain of the paper and the technology used were different.

Overall, this study suggests that BERTopic should not be viewed as a replacement for human thematic analysis, but rather as a complementary exploratory tool. When combined with human interpretation,

it can make the analysis of large literature corpora more targeted, efficient, and manageable, while retaining the contextual and theoretical judgement necessary to produce meaningful findings.

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Appendices:

Appendix A:

1. A full list of 169 CST papers used in the corpus can be found in csv. file= BERT_topics_new
2. Duplicate paper (81) in Survey Paper:

Year	Publication
2024	[7, 10, 11, 17, 25, 29, 31–33, 36, 46, 57, 60, 65, 71, 79, 81, 81, 82, 92, 94, 97, 115, 127, 137, 140, 146, 152, 155, 158, 163, 172, 174–176, 184, 187, 189, 190, 197, 201, 204, 208, 213, 217, 219, 221, 223–225, 228, 229]
2023	[9, 23, 30, 37, 45, 59, 76, 80, 83, 85, 88, 91, 101, 110, 117, 133, 145, 147, 151, 171, 185, 188, 191, 193, 194, 199, 205, 210, 215, 216, 226]
2022	[2, 53, 55, 67, 102, 116, 156, 159, 177, 180, 183, 222]
2021	[35, 52, 63, 84, 86, 90, 95, 118, 119, 123, 136, 160, 178, 212]
2020	[4, 5, 13, 19, 22, 47, 106, 107, 114, 135, 141, 164, 165, 179, 186, 220, 231, 232]
2019	[1, 3, 12, 70, 105, 125, 128, 132, 134, 150, 167, 202, 203]
2018	[64, 121, 139, 192, 206]
2017	[8, 21, 24, 49, 54, 69, 72, 74, 98, 126, 138, 148, 166, 200, 207]
2016	[26, 42, 129, 169, 182]
2015	[48, 50, 73, 99, 100, 162, 168, 170]

Table 1: List of included papers (sorted by year).

3. BERTopic key words Representation:

(csv. file= BERTopic_info_keywords_new)

Topic	Count	Name	Representation
-1	32	-1_llm_ml_comic_risk	['llm', 'ml', 'comic', 'risk', 'template', 'edit', 'child', 'story', 'toolkit', 'chatgpt']
0	42	0_board_analogy_mood board_mood	['board', 'analogy', 'mood board', 'mood', 'generative ai', 'fig', 'design space', 'generate image', 'adjective', 'conceptual']
1	22	1_vr_sound_musical_visualization	['vr', 'sound', 'musical', 'visualization', 'adaptive', 'spatial', 'rice', 'factor', 'ar', 've']
2	16	2_crowd_ideator_idea generation_partner	['crowd', 'ideator', 'idea generation', 'partner', 'worker', 'generate idea', 'chat', 'difference', 'facilitator', 'mind']
3	13	3_character_story_viewer_narrative	['character', 'story', 'viewer', 'narrative', 'chart', 'writer', 'streamer', 'stream', 'live', 'news']
4	13	4_writing_writer_metaphor_journalist	['writing', 'writer', 'metaphor', 'journalist', 'story', 'variation', 'llm', 'inject', 'spark', 'language model']
5	8	5_robot_child_social robot_drawing	['robot', 'child', 'social robot', 'drawing', 'non creative', 'creative condition', 'jibo', 'child creativity', 'measure', 'story']
6	7	6_palette_painting_paint_colorization	['palette', 'painting', 'paint', 'colorization', 'symbol', 'brush', 'drawing', 'gradient', 'novice', 'chinese']
7	6	7_child_story_avatar_scratch	['child', 'story', 'avatar', 'scratch', 'storytelling', 'collage', 'programming', 'character', 'drawing', 'visual storytelling']
8	6	8_co creative_human ai_agent_creative system	['co creative', 'human ai', 'agent', 'creative system', 'drawing', 'robot', 'ai agent', 'wizard', 'reframer', 'initiative']
9	4	9_animation_puppet_rig_character	['animation', 'puppet', 'rig', 'character', 'template', 'shadow', 'pose', 'shadow puppet', 'cinematic', 'secondary']

Appendix B: Summary of attached files

1. **Corpus with full text, DOI, retrieval source, date etc:**

- Final_cst_corpus_with_fulltext_final.csv , this files is used for preprocessing

2. **Embedding files (uploaded in final codebook BERTopic:**

Clean text for BERT model:

- Corpus_preprocessed.csv

3. **Document embeddings for transformer model:**

- document_embeddings-2.npy

4. **BERTopic results:**

- BERT_topics_new.csv
- BERTopic_info_keywords_new.csv

5. **Code book .ipynb files:**

1. 1_Extracting_full_text.ipynb
2. 2_Preprocessing.ipynb
3. 3_Final_BERTopic.ipynb